

SUDARSHAN ANAND

Atlanta, GA (willing to relocate) | asudarshan14@gmail.com | +1 (404) 453-5664 |
Linkedin: www.linkedin.com/in/sanand315/ | Portfolio: sudarshanand14.vercel.app

F-1 OPT (2026–2029)

SUMMARY

NSF-EMBS-Google NextGen Scholar and AI researcher focused on interpretable, multimodal ML systems across healthcare and foundation models. Bridges mathematical rigor with scalable, production-ready AI systems under real-world data constraints, with emphasis on robustness, uncertainty, and evaluation-driven modeling.

EDUCATION

Georgia Institute of Technology Atlanta, GA, USA
MS Computational Science and Engineering (CGPA: 4.0/4.0) Aug 2024 – May 2026
Relevant Coursework: Data Science for Epidemiology, Machine Learning for Graphs, Industrial Math, Numerical Linear Algebra

Birla Institute of Technology and Science (BITS Pilani) Pilani, Rajasthan, India
MSc. Mathematics and B.E. Computer Science (CGPA: 9.4/10) Aug 2019 – Aug 2024

HONORS & AWARDS

IEEE BHI 2025 Data Challenge Competition Winner Oct 2025

- Led development of an **AI/ML-based depression risk prediction system**, contributing to a winning solution in the IEEE-sponsored competition (supported by NSF and Google)

IEEE BHI 2025 Young Professional NextGen Scholar (NSF, EMBS, Google Sponsored) Sep 2025

- Selected for a **competitive NSF/Google-sponsored scholarship** recognizing high-impact biomedical AI research; invited to present at IEEE BHI 2025

SKILLS

- Data:** Python, SQL, Pandas, NumPy, SciPy
- Machine Learning / AI:** PyTorch (distributed training, mixed precision), Scikit-learn, Transformers, self-supervised learning (SimCLR), multimodal learning, RAG systems, fine-tuning (SFT), uncertainty-aware modeling
- Systems & Infrastructure:** HPC workflows (SLURM-based scheduling), distributed training pipelines, FastAPI, GCP (Cloud Run), PostgreSQL, Pinecone, Next.js
- Analysis & Visualization:** Statistical analysis, benchmarking, Seaborn, Plotly

PROJECTS

DRiP: White-box framework for Depression Risk Prediction Oct 2025
IEEE-EMBS International Conference on Biomedical Health Informatics (BHI) 2025 Atlanta, GA, USA

- Built **interpretable longitudinal depression risk model for sparse clinical data (n=200)**, enabling feature-level temporal attribution under constrained model complexity
- Developed **stacked classical ML ensemble** achieving $R^2 = 0.80$ (vs. ~ 0.40 baseline) on patient-level held-out real data under 10-fold cross-validation
- Evaluated distribution-matched synthetic augmentation (~ 600 samples) and performed post-hoc temporal attribution to extract stable risk and treatment-response signals.

Multimodal Progression Tracking of Neurodegenerative disorders Aug 2025 – present
Georgia Institute of Technology Atlanta, GA, USA

- Built a multimodal pipeline for Parkinson's disease progression modeling, fine-tuning a **SimCLR-based self-supervised model (ResNet-18 backbone)** on 3-class staging (control vs prodromal vs Parkinson's) using structural MRI with demographic fusion (age, gender) for holistic patient information.
- Evaluated a clinically ambiguous classification task with **weak structural separability between early disease stages**, using class-wise F1 (macro F1 ≈ 0.60) to better reflect overlapping phenotypes instead of accuracy.
- Scaled **distributed PyTorch training on HPC clusters using SLURM** for large longitudinal neuroimaging workloads.
- Developed and debugged HPC infrastructure workflows using **LLM-assisted engineering (Claude-based tooling)** to accelerate deployment and scaling of MRI processing pipelines.

Investigating importance of Patient Metadata in Public Health May 2025 – present
Edith Cowan University Western Australia, Australia

- Led analysis of **1.2M unstructured COVID-19 patient records** to evaluate patient metadata utility for modeling host-pathogen interactions in resource-constrained public health settings
- Quantified extreme data limitations, identifying a **94% metadata sparsity rate**, identifying missing metadata as a primary bottleneck for reliable resource planning and clinical interventions during pandemic response.
- Used **state-of-the-art LLMs (Gemma-2, MedGemma-1.5)** as a controlled probe to evaluate robustness under degraded clinical context and missing information regimes.

Samay: Time-series Foundational Models Library Jan 2025 – May 2025

- Developed [Samay](#), an open-source library **unifying 10 time-series foundation models** for forecasting, classification, and anomaly detection, enabling standardized inference and evaluation workflows (**34 stars, 7 forks**)
- Developed modular **PyTorch-based inference and fine-tuning pipelines (PyTorch 2.0+)**, enabling zero-shot evaluation and task-specific adaptation of pretrained models across diverse time-series formats.
- Designed and executed large-scale benchmarking against the [GiftEval benchmark](#), validating model performance and scalability across unified evaluation protocols for time-series foundation models.

PROFESSIONAL EXPERIENCE

AI Product Development Intern

Jun 2025 – Aug 2025

Rezolve.ai

Dublin, CA, USA [Remote]

- Built an enterprise **Retrieval-Augmented Generation (RAG) search MVP** using FastAPI, Pinecone, and PostgreSQL with conversational memory, enabling policy retrieval in **<20s latency** for enterprise knowledge workflows.
- Developed a **prefix-tree (trie)-based autocomplete system** and automated global data pipeline updates using **GCP Cloud Run**.
- Designed prototype for an **Agentic AI ChatOps system (Slack/Teams)** for IT incident response, automating alert triaging and runbook generation to streamline incident workflows.

AI Scientist Intern

Jan 2024 – Jun 2024

Qure.ai Technologies Pvt. Ltd.

Bangalore, Karnataka, India

- Improved a clinical lung nodule detection system, increasing **Pearson correlation with radiologist assessments by ~45% relative to baseline models**, enhancing reliability for early cancer screening workflows.
- Benchmarked image registration performance for longitudinal CT-based lung nodule tracking, evaluating **temporal alignment consistency across consecutive scans** for clinical follow-up analysis.

SELECTED PUBLICATIONS

- **Foundation Models / ML Systems**
 - H. Kamarthi, S. Li, **S. Anand**, S. Das, A. Ramesvaran, A. Bhalerao and B.A. Prakash, “*Samay: A Unified Pipeline Framework for Time-Series Foundation Models*” **VLDB 2026** (Demo track , submitted)
- **Public Health / Data-Centric AI**
 - **S. Anand**, M. Lee, and S. S. Vasan, “*Metadata makes Big Data useful for AI/ML-ready, equitable pandemic response*” **IEEE-EMBS BHI 2025 (Poster)**
 - **S. Anand**, M. Lee, and S. S. Vasan, “*Why de-identified patient meta-data matters during pandemics*” **Scotland's Health Research and Innovation Conference 2025 (Poster)**
- **Optimization / Systems (Deployed Academic Product)**
 - Shiksha, **Anand S**, Shekhawat K, Agrawal K. *Automated generation of circulations within a floorplan*. **Artificial Intelligence for Engineering Design, Analysis and Manufacturing**. 2025.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug 2025 – present

Georgia Institute of Technology (CSE Algorithms (Aug-Dec '25); Numerical Linear Algebra (Jan '26-present))

Atlanta, GA, USA

- Facilitate instruction for **180 students across Algorithms and Numerical Linear Algebra**, providing targeted conceptual reinforcement and assignment feedback